Practical 6

Considered there are N philosophers seated around a circular table with one chopstick between each pair of philosophers. There is one chopstick between each philosopher. A philosopher may eat if he can pick up the two chopsticks adjacent to him. one chopstick may be picked up by any one of its adjacent followers but not both.

Write a program to solve the problem using process synchronization.

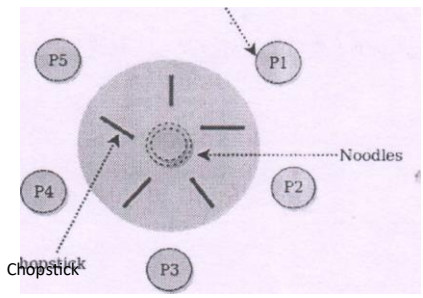

Fig: 1. Dining Philosopher Problem

Code:

main.c

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```c
#include <stdio.h>
#define N 5
int main()
{
    int i;

    for(i=0;i<N;i++)
    {
        printf("Philosopher %d is Thinking\n", i+1);
        printf("Philosopher %d takes chopstick %d and %d\n", i+1, i+1, (i+1)%N+1);
        printf("Philosopher %d is Eating\n", i+1);
        printf("Philosopher %d puts down chopstick %d and %d\n", i+1, i+1, (i+1)%N+1);
        printf("\n");
    }

    return 0;
}
```

Output:

Output

```
Philosopher 1 is Thinking
Philosopher 1 takes chopstick 1 and 2
Philosopher 1 is Eating
Philosopher 1 puts down chopstick 1 and 2

Philosopher 2 is Thinking
Philosopher 2 takes chopstick 2 and 3
Philosopher 2 is Eating
Philosopher 2 puts down chopstick 2 and 3

Philosopher 3 is Thinking
Philosopher 3 takes chopstick 3 and 4
Philosopher 3 is Eating
Philosopher 3 puts down chopstick 3 and 4

Philosopher 4 is Thinking
Philosopher 4 takes chopstick 4 and 5
Philosopher 4 is Eating
Philosopher 4 puts down chopstick 4 and 5

Philosopher 5 is Thinking
Philosopher 5 takes chopstick 5 and 1
Philosopher 5 is Eating
Philosopher 5 puts down chopstick 5 and 1



=== Code Execution Successful ===
```